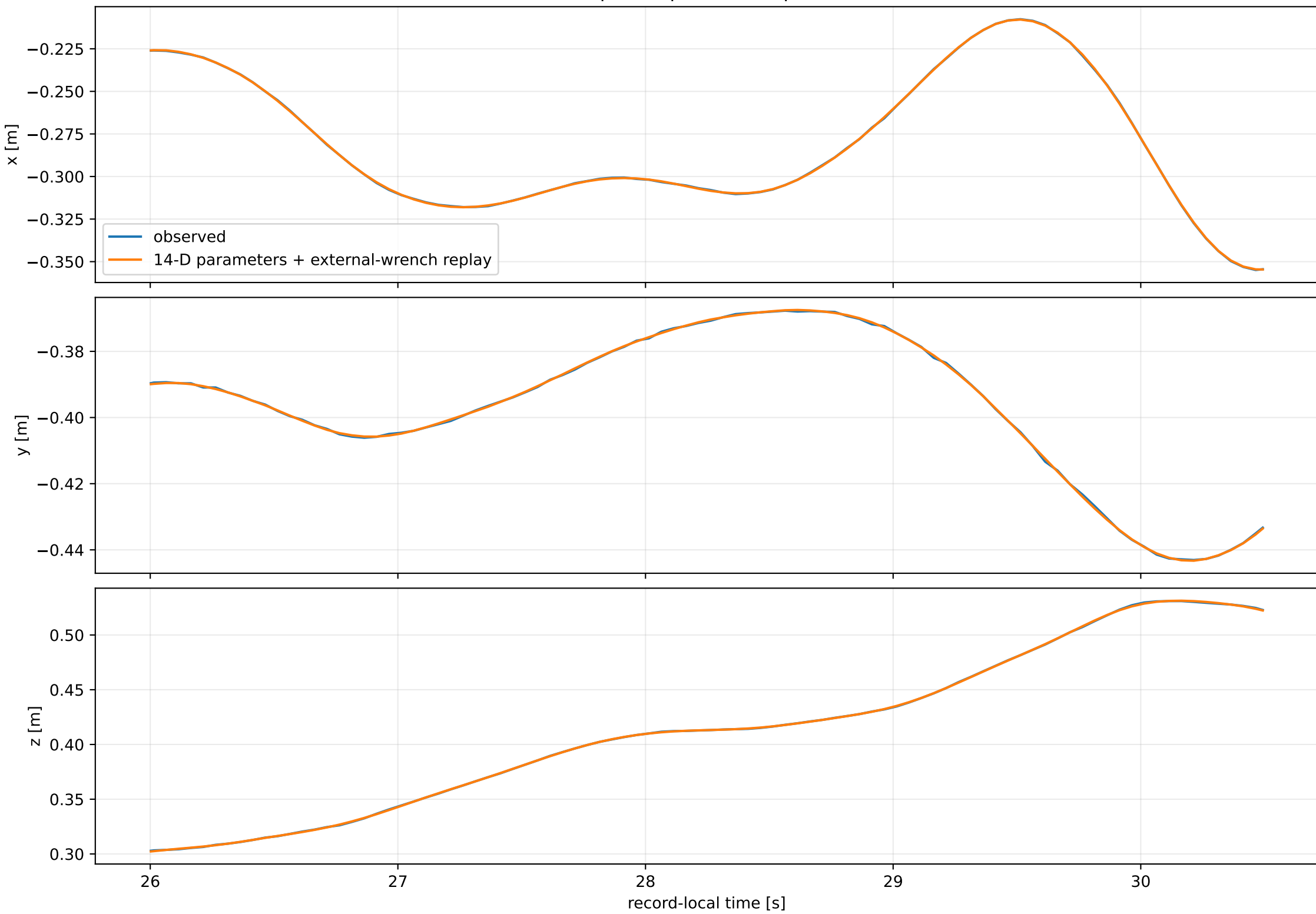
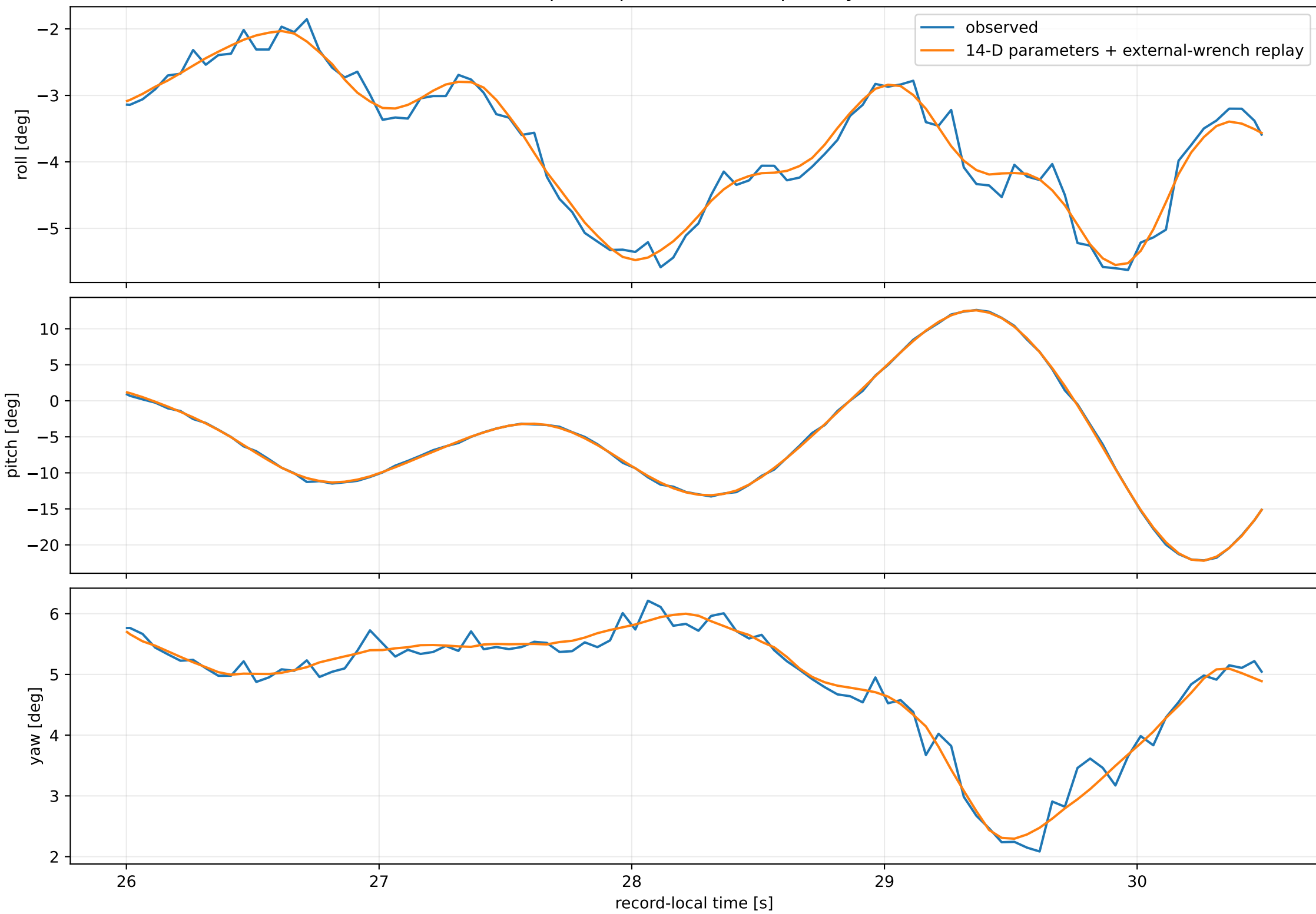


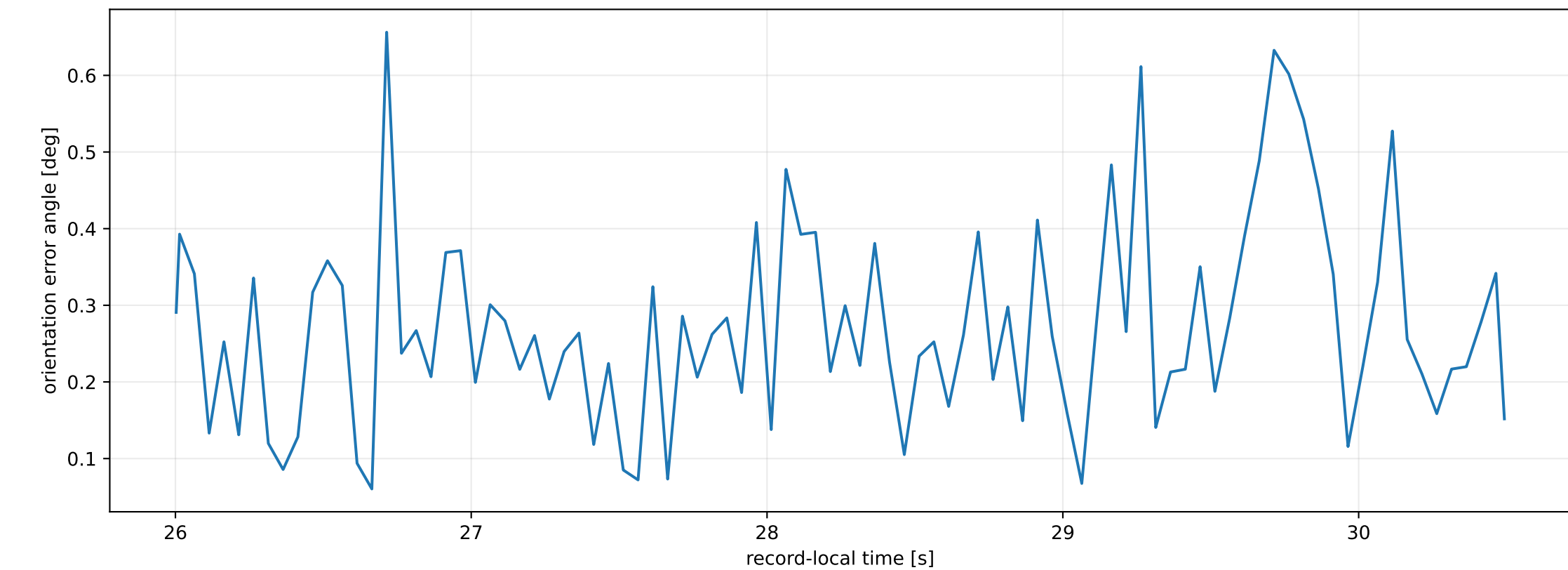
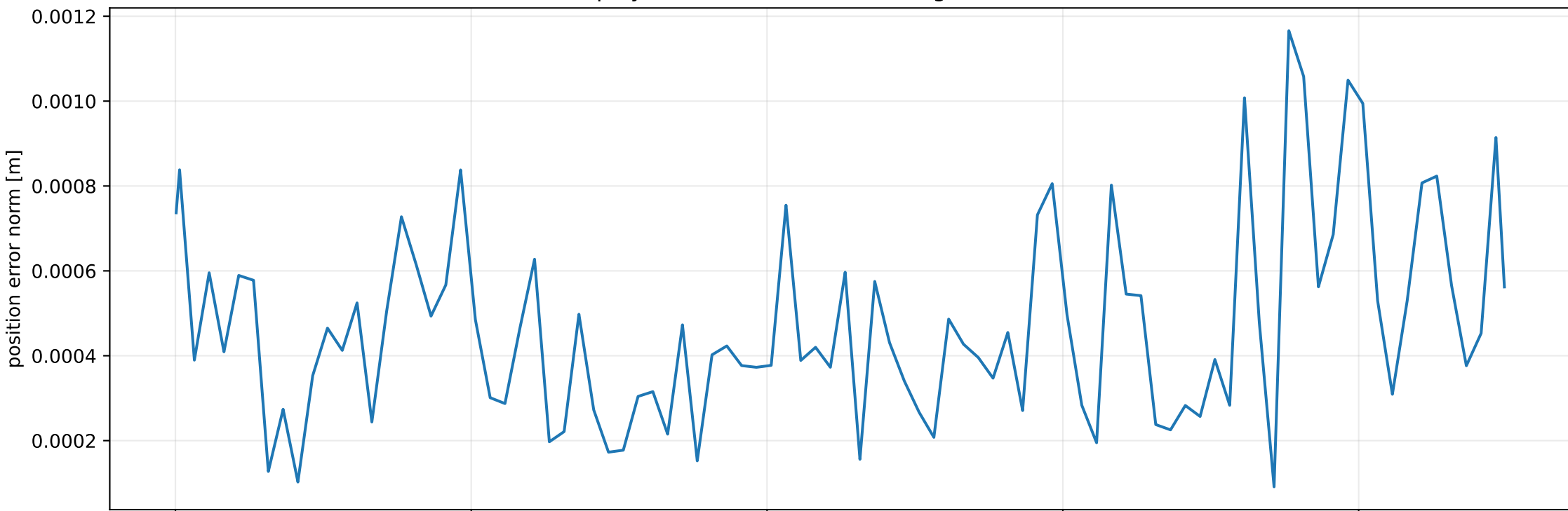
Fine pose reproduction: position



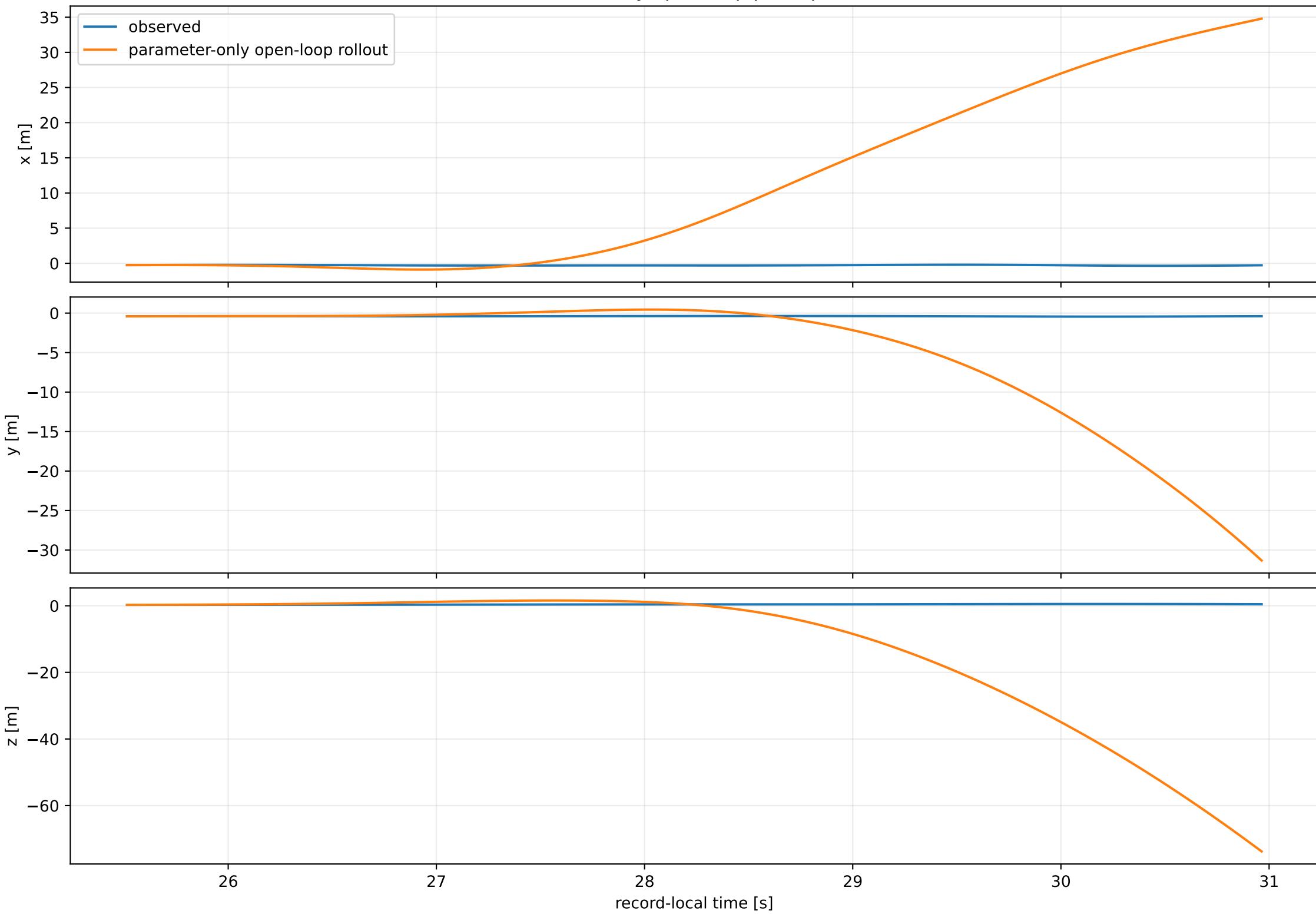
Fine pose reproduction: roll / pitch / yaw



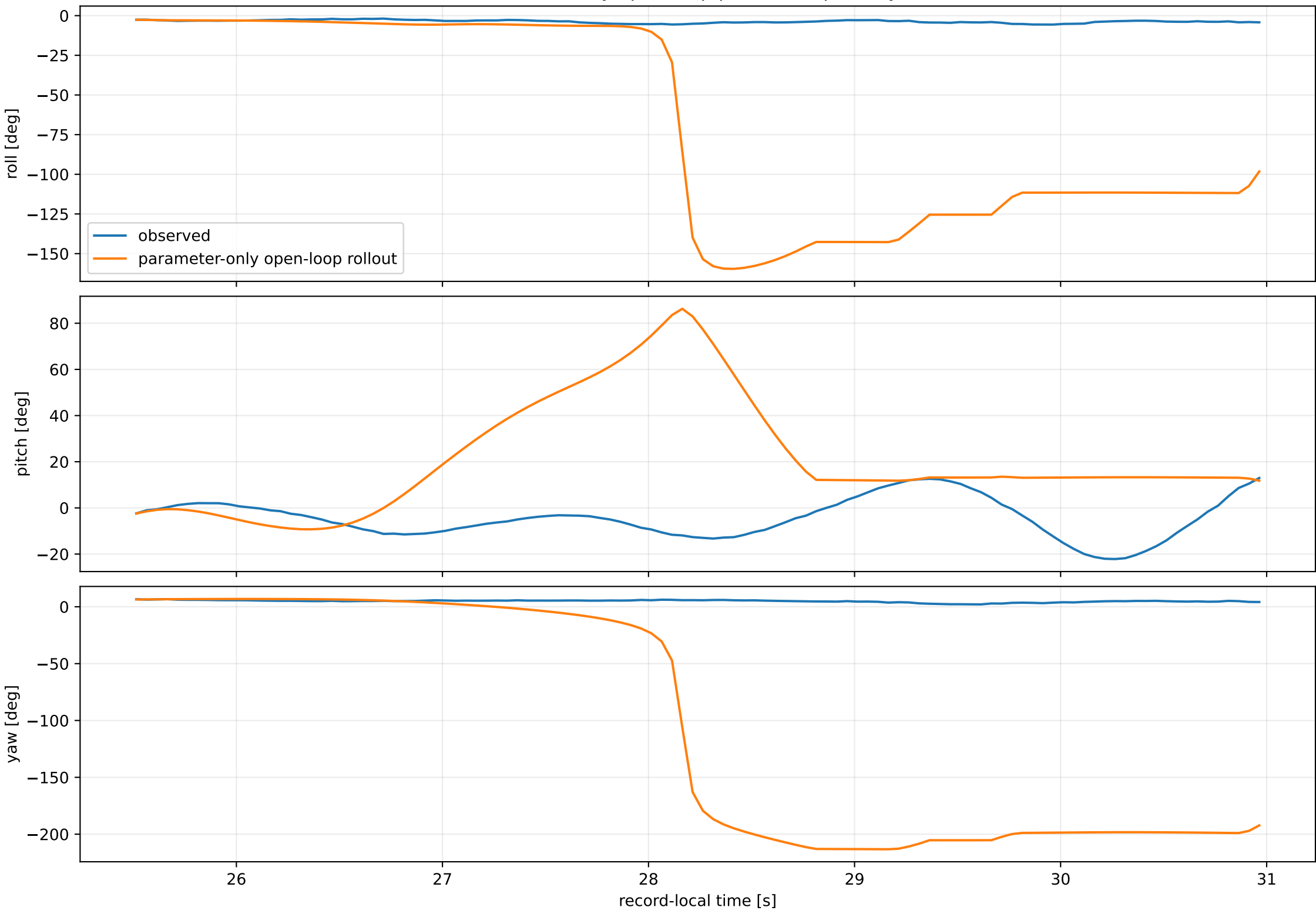
Fine replay error after the 14-D change — lower is better



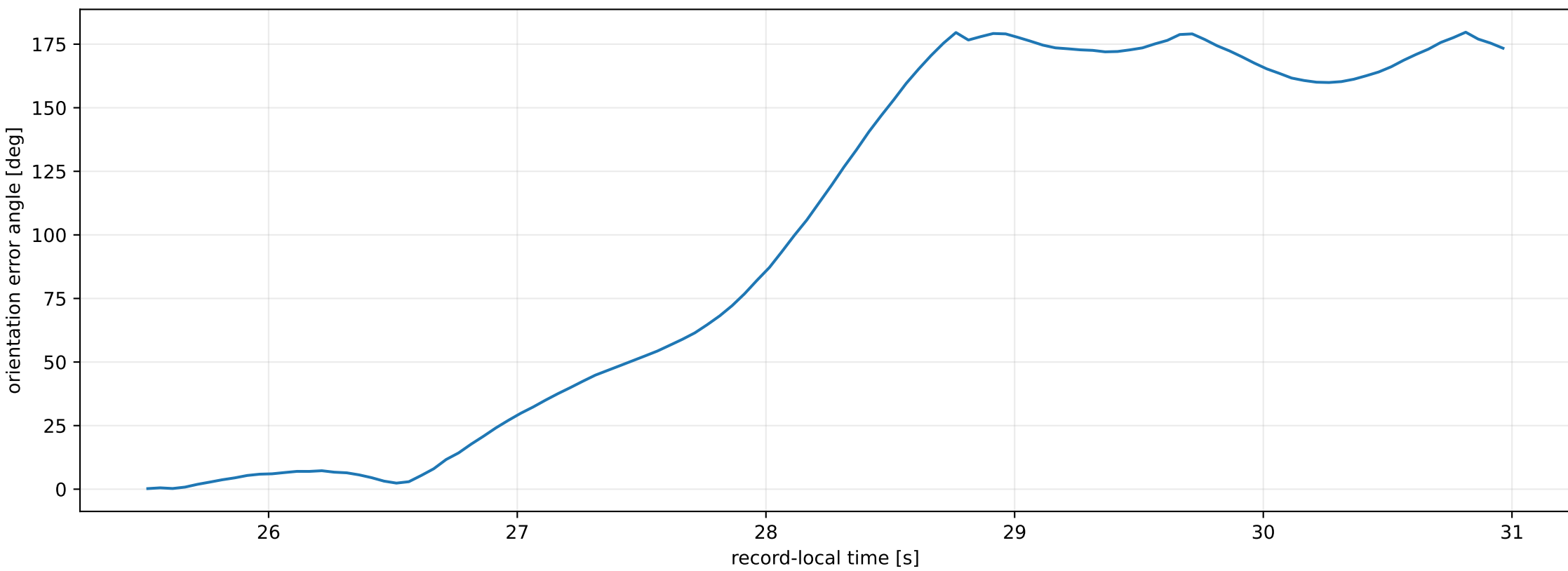
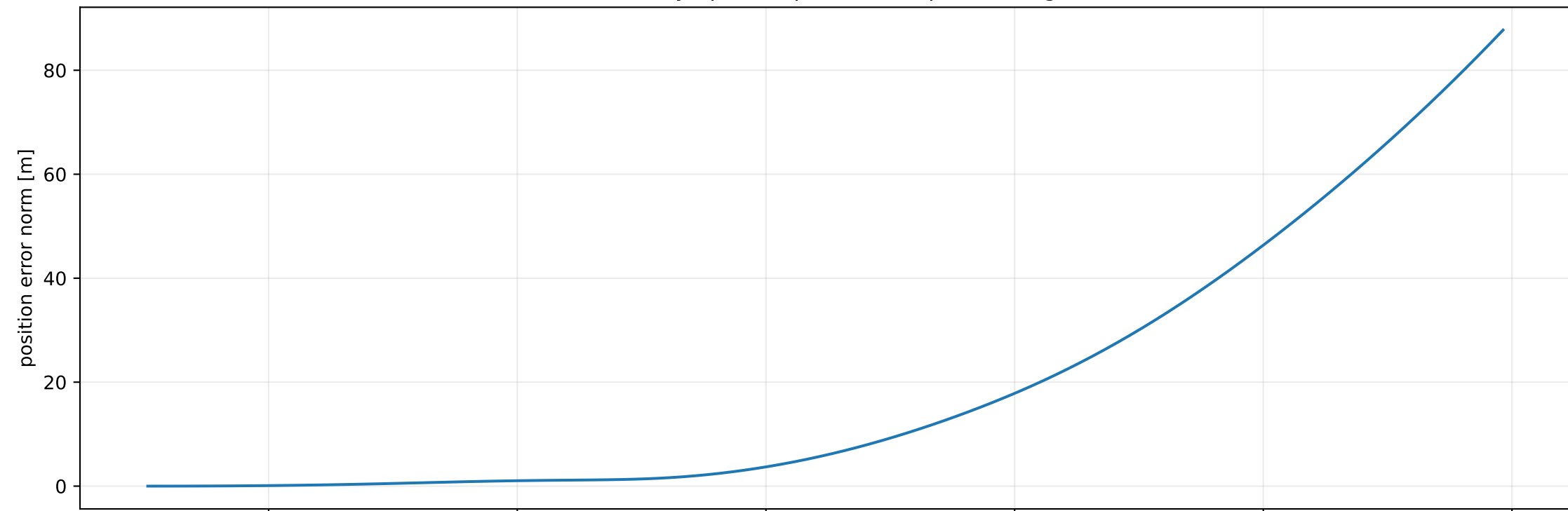
Parameter-only open-loop pose: position



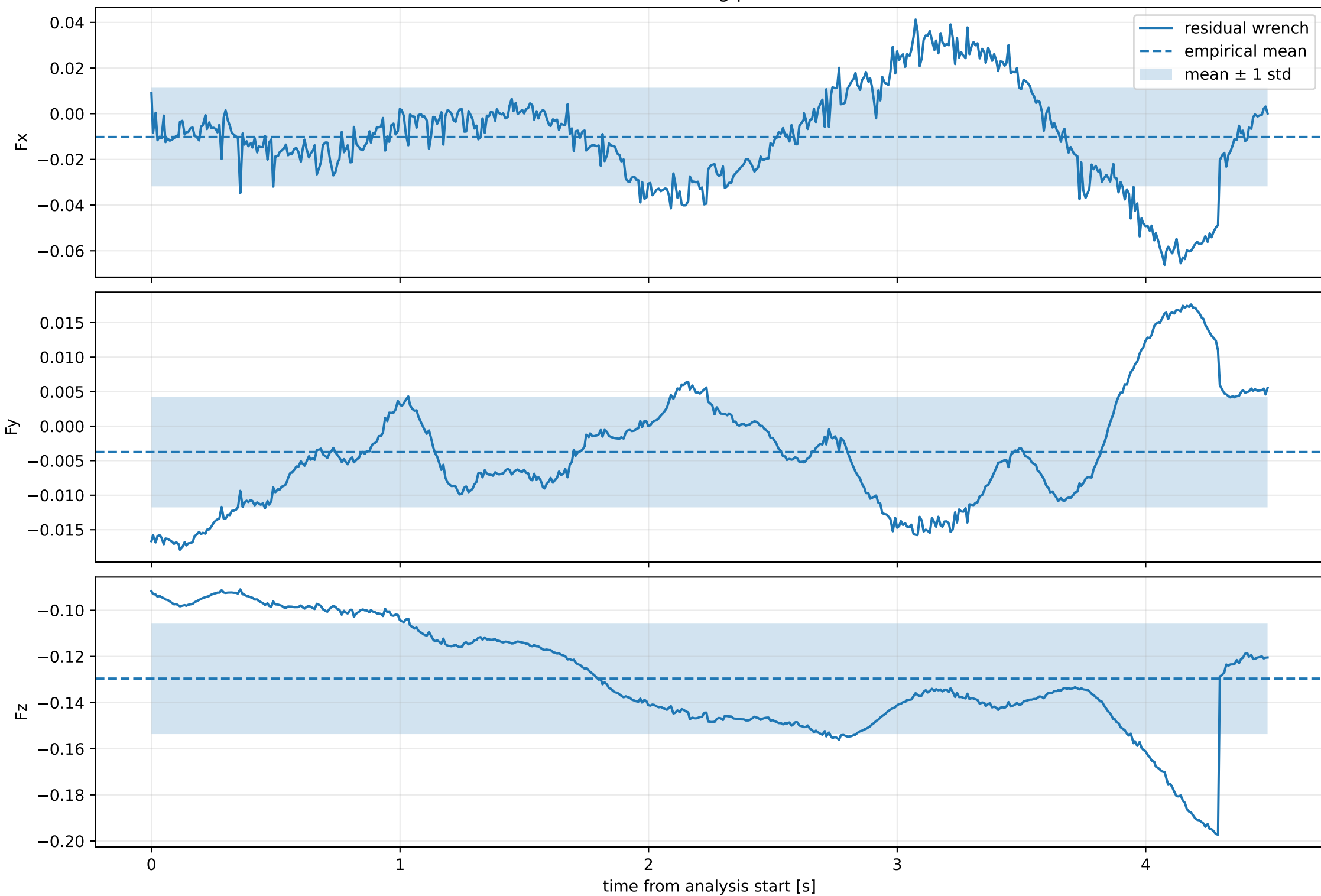
Parameter-only open-loop pose: roll / pitch / yaw



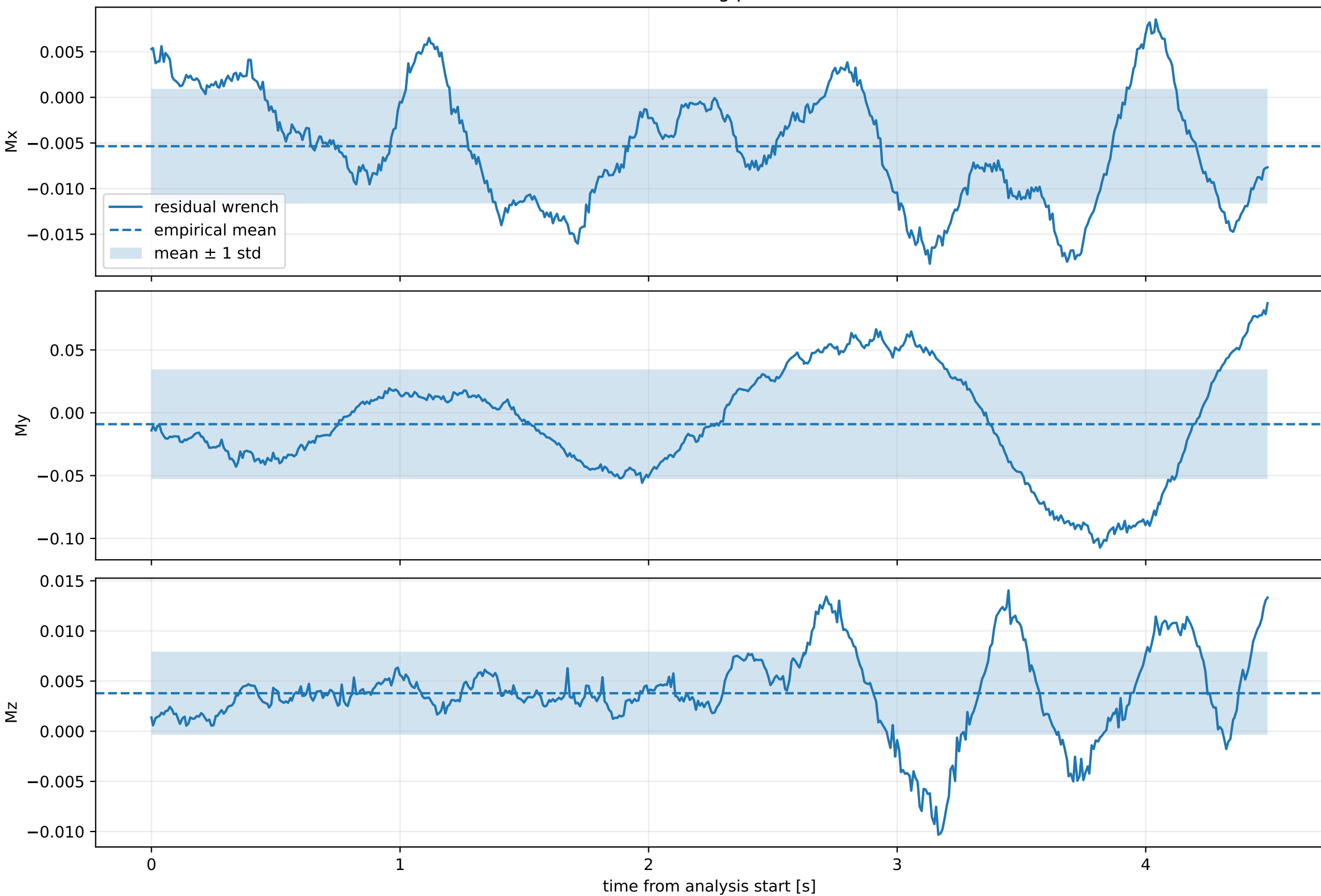
Parameter-only open-loop rollout (separate diagnostic)



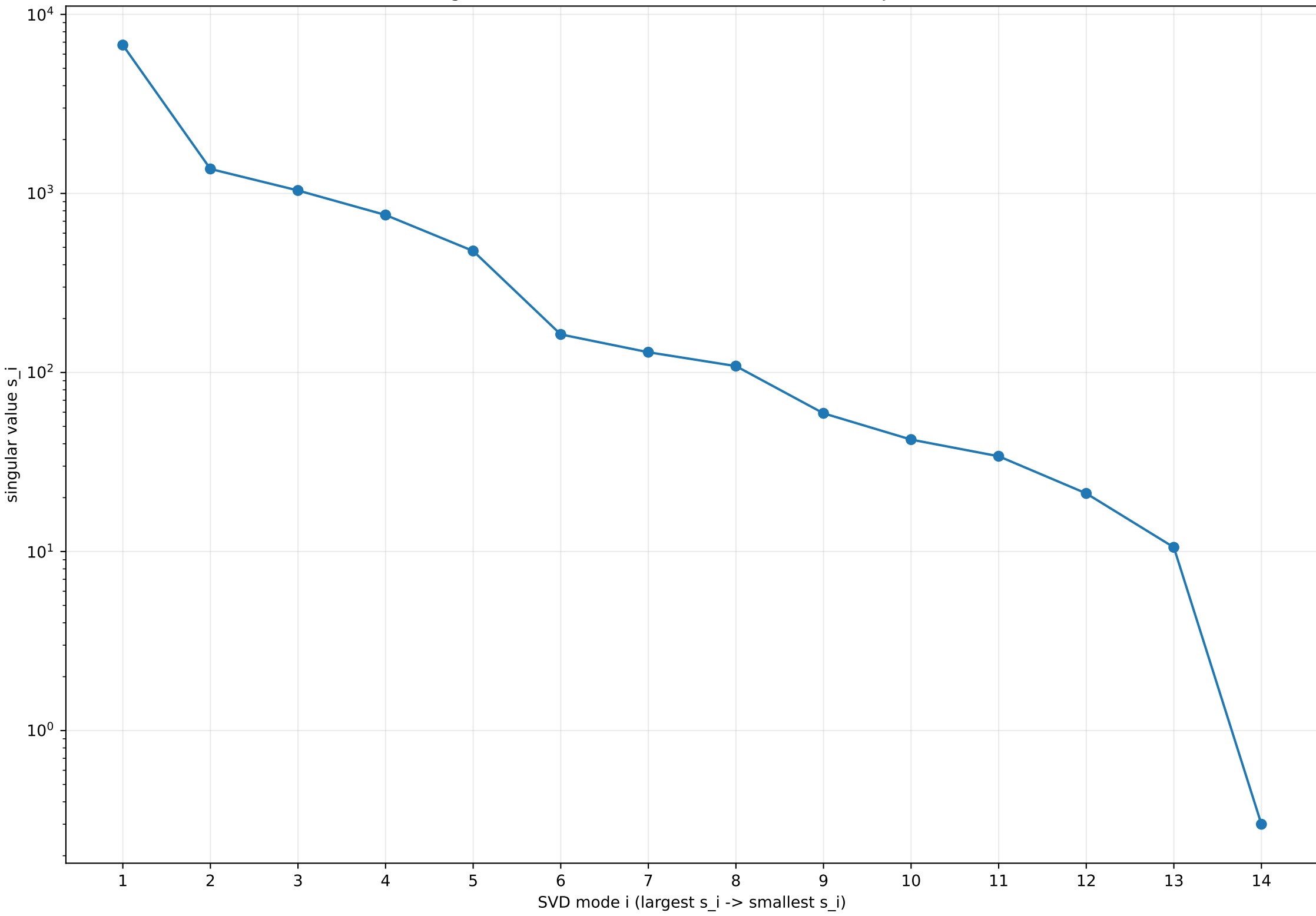
Dimensionless residual force: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



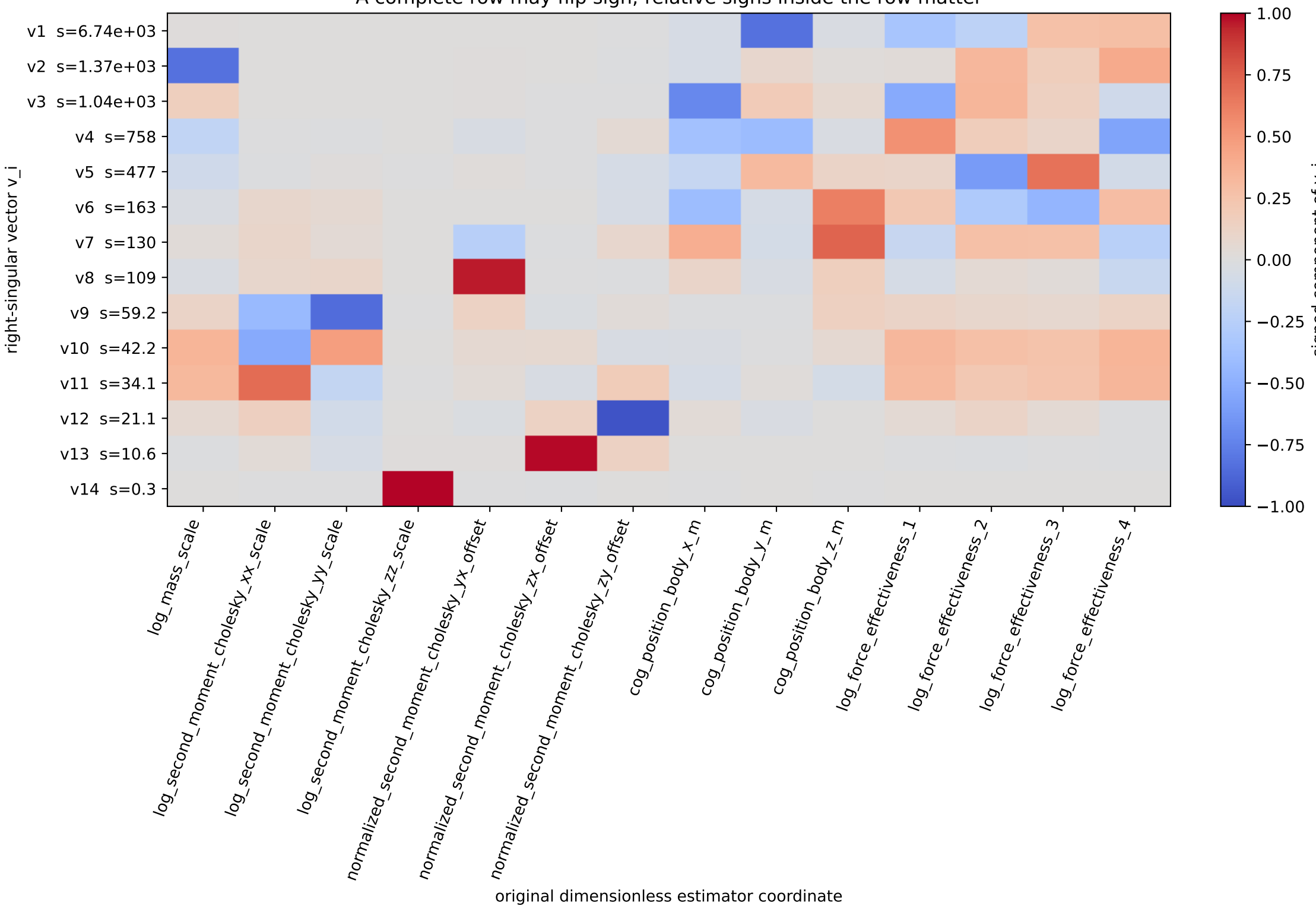
Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36648098134

inertia J [kg m²]

[0.065002586 6.1448417e-06 1.8187244e-05]
[6.1448417e-06 0.065038584 6.7095457e-07]
[1.8187244e-05 6.7095457e-07 0.12899228]

principal moments [kg m²]

[0.065001561, 0.065039603, 0.12899229]

CoG position in body frame [m]

[-0.002734852, 0.0003877356, 0.0095455595]

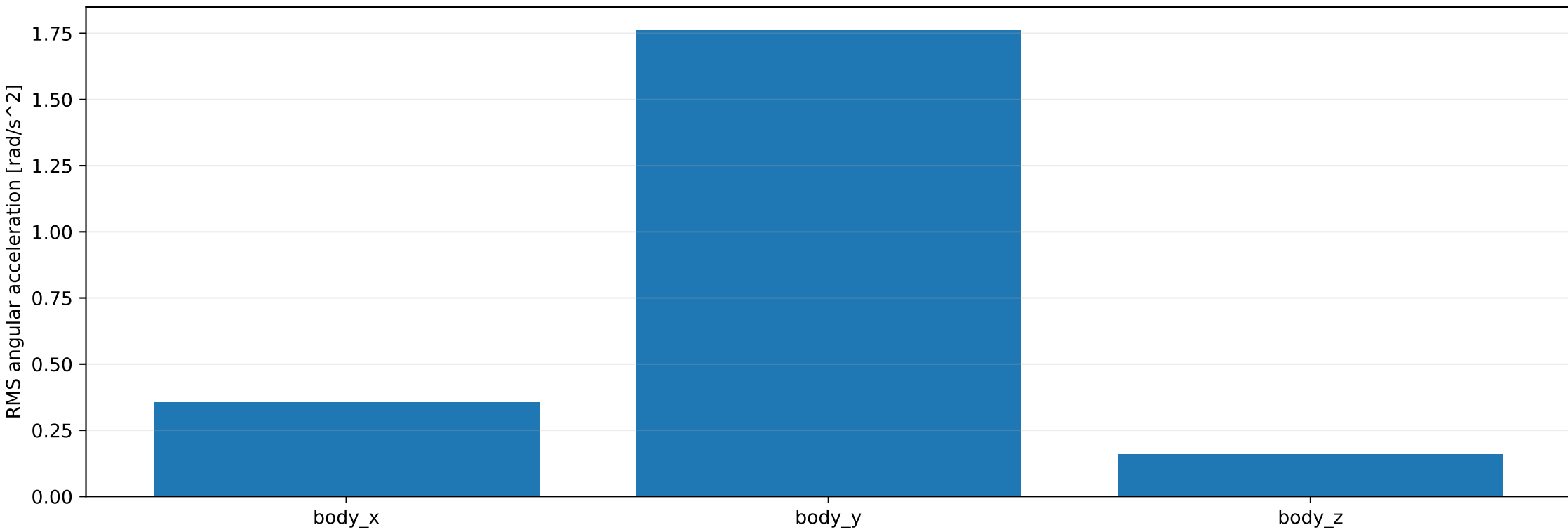
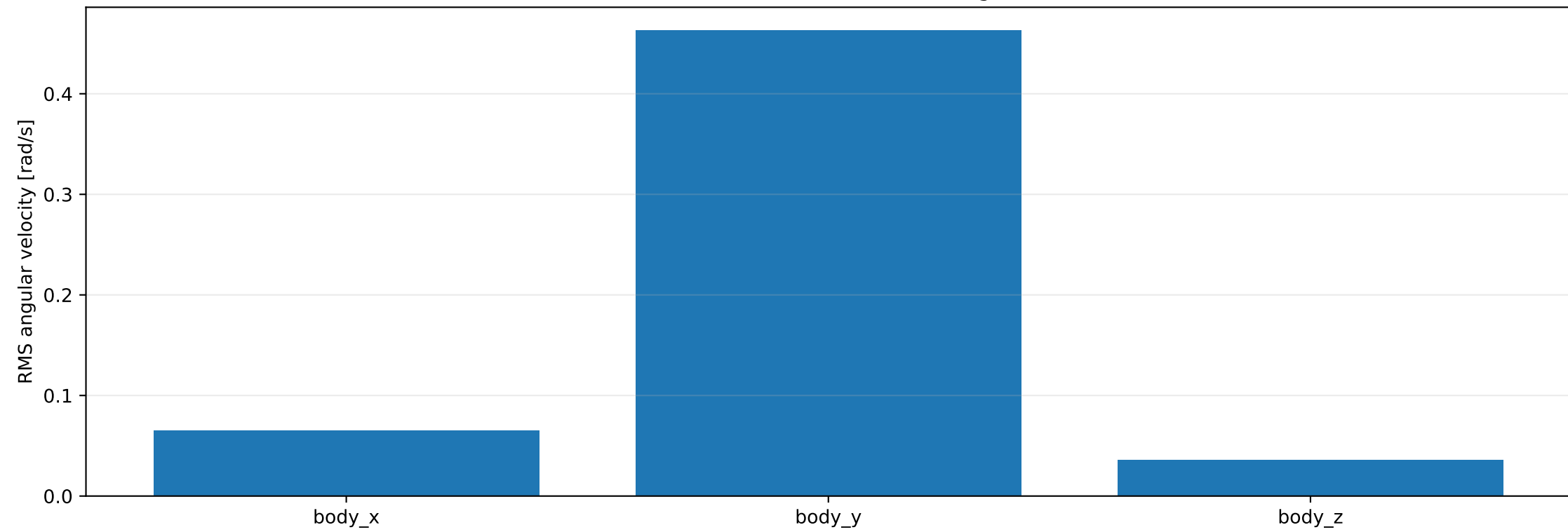
rotor force effectiveness

[0.96022656, 0.97096555, 0.9650767, 0.95485367]

selected delay [s]

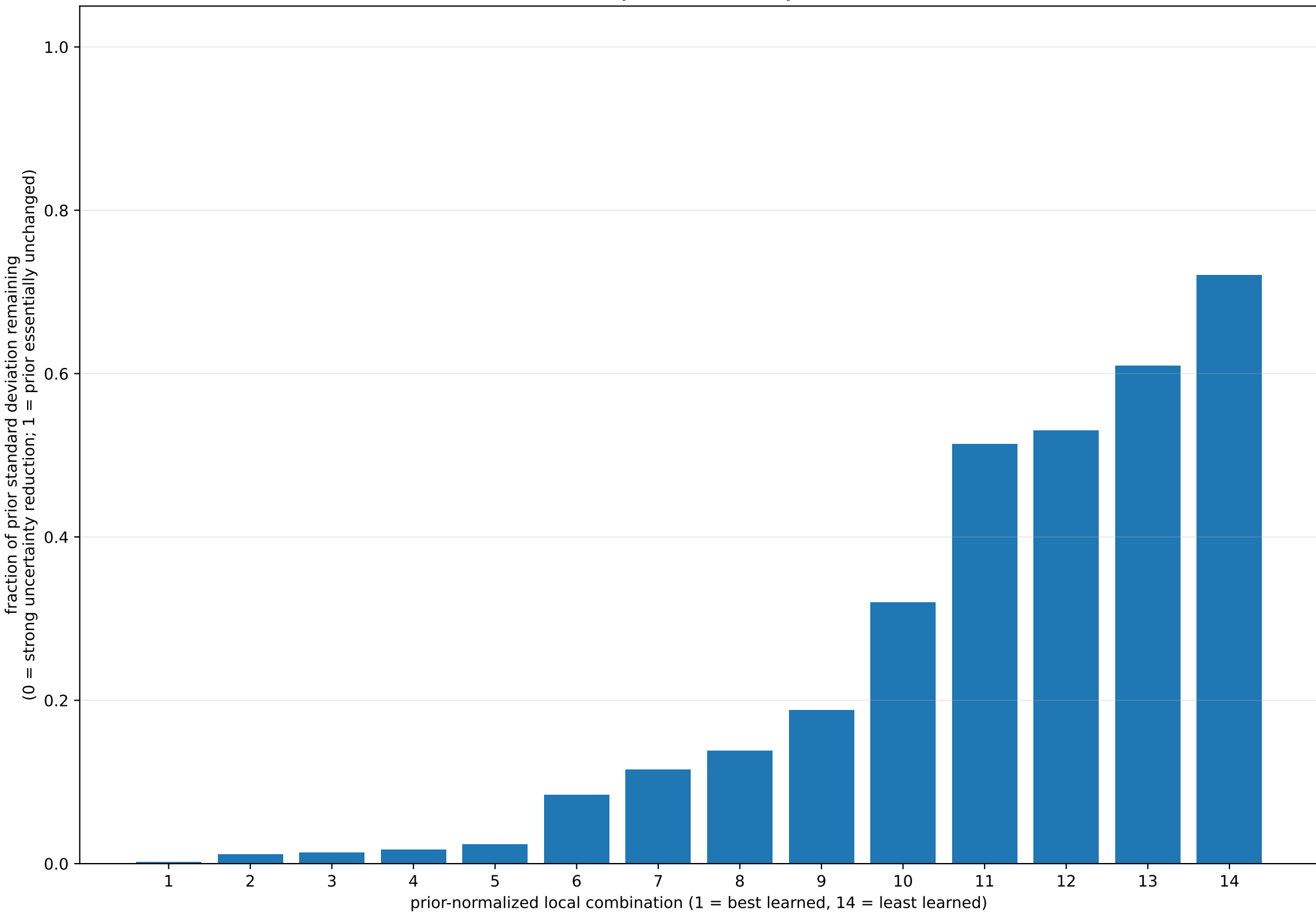
0.012308586022

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

